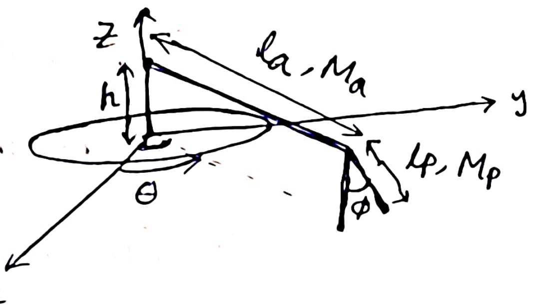
~~Pend~~ Inverted Pendulum



1. For the centre rod,
Let, moment $= J$, height $= h$
mass $= M_c$

$K_c = \frac{1}{2} J \cdot \dot{\theta}^2 \qquad V_c = M_c\, g \cdot \frac{h}{2}$

(Assume mass is uniformly distributed)

2. For the axis rod, length $= l_a$, mass $= M_a$
mass density $= \frac{M_a}{l_a}$

The coordinate of a point r_a distance from pivot on the rod is,

$$\begin{cases} x_a = r_a \cos\theta & \Rightarrow \dot{x}_a = -r_a \sin\theta\, \dot{\theta} \\ y_a = r_a \sin\theta & \Rightarrow \dot{y}_a = r_a \cos\theta\, \dot{\theta} \\ z_a = h & \Rightarrow \dot{z}_a = 0 \end{cases}$$

velocity of that point $= v_a$

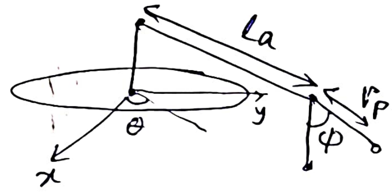
$$v_a^2 = \dot{x}_a^2 + \dot{y}_a^2 + \dot{z}_a^2 = r_a^2 \dot{\theta}^2$$

$$2K_a = \int_0^{l_a} r_a^2 \dot{\theta}^2 \cdot \frac{M_a}{l_a}\, dr_a = \frac{M_a}{l_a} \cdot \dot{\theta}^2 \cdot \frac{l_a^3}{3}$$
$$= \frac{1}{3} M_a \cdot l_a^2 \cdot \dot{\theta}^2$$

$$V_a = M_a \cdot g \cdot h$$

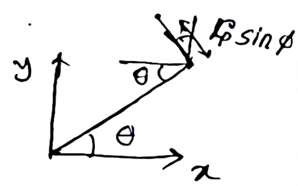
3. For the pendulum rod, length $= l_p$, mass $= M_p$
mass density $= \frac{M_p}{l_p}$

The coordinate of a point r_p distance from pivot on the rod (pendulum)



$$\begin{cases} x_p = l_a \cos\theta - r_p \sin\phi \sin\theta \\ y_p = l_a \sin\theta + r_p \sin\phi \cos\theta \\ z_p = h - r_p \cos\phi \end{cases}$$

top view



$$\dot{x}_p = -l_a \sin\theta\, \dot{\theta} - r_p \sin\phi \cos\theta\, \dot{\theta} - r_p \cos\phi \sin\theta\, \dot{\phi}$$

$$\dot{y}_p = l_a \cos\theta\, \dot{\theta} + r_p \cos\phi \cos\theta\, \dot{\phi} - r_p \sin\phi \sin\theta\, \dot{\theta}$$

$$\dot{z}_p = r_p \sin\phi\, \dot{\phi}$$

$$\cancel{v_p^2 = \dot{x}_p^2 + \dot{y}_p^2 + \dot{z}_p^2} = \cancel{(l_a \sin\theta - r_p \sin\phi \cos\theta)^2 \dot{\theta}^2}$$

$$v_p^2 = \dot{x}_p^2 + \dot{y}_p^2 + \dot{z}_p^2$$

$$= (-l_a \sin\theta - r_p \sin\phi \cos\theta)^2 \dot{\theta}^2 + (r_p \cos\phi \sin\theta)^2 \dot{\phi}^2$$

$$+2(l_a \sin\theta + r_p \sin\phi \cos\theta)(r_p \cos\phi \sin\theta)\dot{\theta}\dot{\phi}$$

$$+ (l_a \cos\theta - r_p \sin\phi \sin\theta)^2 \dot{\theta}^2 + (r_p \cos\phi \cos\theta)^2 \dot{\phi}^2$$

$$+2(l_a \cos\theta - r_p \sin\phi \cdot \sin\theta)(r_p \cos\phi \cos\theta)\dot{\theta}\dot{\phi}$$

$$+(r_p \sin\phi)^2 \dot{\phi}^2$$

$$= \dot{\theta}^2 (l_a^2 \sin^2\theta + r_p^2 \sin^2\phi \cos^2\theta + l_a^2 \cos^2\theta + r_p^2 \sin^2\phi \sin^2\theta)$$

$$+ \dot{\phi}^2 (r_p^2 \cos^2\phi \sin^2\theta + r_p^2 \cos^2\phi \cos^2\theta + r_p^2 \sin^2\phi)$$

$$+2\dot{\theta}\dot{\phi}(r_p \cos\phi)(l_a \sin^2\theta + r_p \sin\theta \sin\phi \cos\theta + l_a \cos^2\theta - r_p \sin\theta \sin\phi \cos\theta)$$

$$= \dot{\theta}^2 (l_a^2 + r_p^2 \sin^2\phi) + \dot{\phi}^2 (r_p^2) + 2 l_a \dot{\theta}\dot{\phi} r_p \cos\phi$$

$$2T_p = \int_0^{l_p} v_p^2 \frac{M_p}{l_p} dr_p$$

$$= \frac{M_p}{l_p} \int_0^{l_p} \left\{ (l_a^2 + r_p^2 \sin^2\phi)\dot{\theta}^2 + \dot{\phi}^2 r_p^2 + 2 l_a r_p \dot{\theta}\dot{\phi} \cos\phi \right\} dr_p$$

$$= \frac{M_p}{l_p} \left[\dot{\theta}^2 \left(l_a^2 l_p + \frac{l_p^3}{3} \sin^2\phi\right) \dot{\theta}^2 + \frac{l_p^3}{3} \dot{\phi}^2 + l_a \cdot \frac{l_p^2}{2} \cdot 2 \cdot \dot{\theta}\dot{\phi} \cos\phi \right]$$

$$= \dot{\theta}^2 \left[M_p l_a^2 + \frac{1}{3} M_p l_p^2 \sin^2\phi \right] + \frac{1}{3} M_p l_p^2 \dot{\phi}^2 + l_a \cdot l_p \cdot M_p \cdot \dot{\theta}\dot{\phi} \cos\phi$$

$$V_p = M_p g \cdot \left(h - \frac{l_p}{2} \cos\phi\right)$$

$$\mathcal{L} = K - V$$

$$= \frac{1}{2} J \dot{\theta}^2 - M_c \cdot g \cdot \frac{h}{2} + \frac{1}{2}\left(\frac{1}{3} M_a l_a^2 \dot{\theta}^2\right) - M_a g h$$
$$+ \frac{1}{2}\left(M_p l_a^2 + \frac{1}{3} M_p l_p^2 \sin^2\phi\right)\dot{\theta}^2 + \frac{1}{2}\left(\frac{1}{3} M_p l_p^2 \dot{\phi}^2\right)$$
$$+ \frac{1}{2}\left(l_a l_p M_p \dot{\theta}\dot{\phi} \cos\phi\right) - \cancel{M_p} M_p g \left(h - \frac{l_p}{2}\cos\phi\right)$$

$$= \dot{\theta}^2\left(\frac{1}{2} J + \frac{1}{6} M_a l_a^2 + \frac{1}{2} M_p l_a^2 + \frac{1}{6} M_p l_p^2 \sin^2\phi\right)$$
$$+ \dot{\phi}^2\left(\frac{1}{6} M_p l_p^2\right) + \dot{\theta}\dot{\phi}\left(\frac{1}{2} M_p l_a l_p \cos\phi\right)$$
$$- M_c g \frac{h}{2} - M_a g h - M_p g h + \cancel{M_p} M_p g \frac{l_p}{2}\cos\phi$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = \dot{\theta}\left(J + \frac{1}{3} M_a l_a^2 + M_p l_a^2 + \frac{1}{3} M_p l_p^2 \sin^2\phi\right)$$
$$+ \dot{\phi}\left(\frac{1}{2} M_p l_a l_p \cos\phi\right)$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}\right) = \ddot{\theta}\left(J + \frac{1}{3} M_a l_a^2 + M_p l_a^2 + \frac{1}{3} M_p l_p^2 \sin^2\phi\right)$$
$$+ \dot{\theta}\left(\frac{1}{3} M_p l_p^2\right) 2\sin\phi \cos\phi \cdot \dot{\phi} + \left(\frac{1}{2} M_p l_a l_p \cos\phi\right)\ddot{\phi}$$
$$- \frac{1}{2} M_p l_a l_p \sin\phi \; \dot{\phi}^2$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi}\left(\frac{1}{3} M_p l_p^2\right) + \dot{\theta}\left(\frac{1}{2} M_p l_a l_p \cos\phi\right)$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}}\right) = \ddot{\phi}\, \frac{1}{3} M_p l_p^2 + \ddot{\theta}\left(\frac{1}{2} M_p l_a l_p \cos\phi\right)$$
$$- \frac{1}{2} M_p l_a l_p \sin\phi \; \dot{\theta}\dot{\phi}$$

$$\frac{\partial \mathcal{L}}{\partial \phi} = \dot{\theta}^2\left(\frac{1}{6} M_p l_p^2 \cdot 2\sin\phi\cos\phi\right)$$
$$- \dot{\theta}\dot{\phi}\, \frac{1}{2} M_p l_a l_p \sin\phi - M_p g \frac{l_p}{2}\sin\phi$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}\right) - \frac{\partial \mathcal{L}}{\partial \theta} = \tau - c\dot{\theta}$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}}\right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0 - b\dot{\phi}$$

friction coeff. of arm $= c$

friction coeff of pendulum $= b$

1) $\ddot{\theta}\left(J+\frac{1}{3}M_a l_a^2 + M_p l_a^2 + \frac{1}{3}M_p l_p^2 \sin^2\phi\right)$
$+\frac{1}{3}M_p l_p^2 \sin(2\phi)\cdot\dot{\theta}\dot{\phi} + \left(\frac{1}{2}M_p l_a l_p \cos\phi\right)\cdot\ddot{\phi}$
$-\frac{1}{2}M_p l_a l_p \sin\phi \;\dot{\phi}^2 = \tau - c\dot{\theta}$

2) $\ddot{\phi}\,\frac{1}{3}M_p l_p^2 + \ddot{\theta}\left(\frac{1}{2}M_p l_a l_p \cos\phi\right) - \frac{1}{2}M_p \cancel{l_a} l_p \sin\phi\,\dot{\theta}\dot{\phi}$
$-\dot{\theta}^2\left(\frac{1}{3}M_p l_p^2 \sin\phi\cos\phi\right) + \frac{1}{2}M_p l_a \cancel{l_p} \sin\phi\,\dot{\theta}\dot{\phi}$
$+M_p\cdot g\cdot\frac{l_p}{2}\sin\phi = 0 - b\dot{\phi}$

Let, $x = \begin{bmatrix}\theta\\ \dot{\theta}\\ \phi\\ \dot{\phi}\end{bmatrix} = \begin{bmatrix}x_1\\ x_2\\ x_3\\ x_4\end{bmatrix}$

(*) $\boxed{\begin{array}{l} J+\frac{1}{3}M_a l_a^2 + M_p l_a^2 = \alpha \\ \frac{1}{3}M_p l_p^2 = \beta,\ \frac{1}{2}M_p g l_p = \delta \\ \frac{1}{2}M_p l_a l_p = \gamma \end{array}}$

$(\alpha+\beta\sin^2\phi)\ddot{\theta} + \beta\sin(2\phi)\dot{\theta}\dot{\phi} + \gamma\cos\phi\,\ddot{\phi}$
$-\gamma\sin\phi\,\dot{\phi}^2 = \tau - c\dot{\theta}$

$\beta\ddot{\phi} + \gamma\cos\phi\,\ddot{\theta} - \beta\sin\phi\cos\phi\cdot\dot{\theta}^2$
$+\frac{1}{2}M_p g l_p \sin\phi = -b\dot{\phi}$

$$\begin{bmatrix}\alpha+\beta\sin^2\phi & \gamma\cos\phi\\ \gamma\cos\phi & \beta\end{bmatrix}\begin{bmatrix}\ddot{\theta}\\ \ddot{\phi}\end{bmatrix} = \begin{bmatrix}-\beta\sin(2\phi)\dot{\theta}\dot{\phi} + \gamma\sin\phi\,\dot{\phi}^2 + \tau - c\dot{\theta}\\ \beta\sin\phi\cos\phi\,\dot{\theta}^2 - \frac{1}{2}M_p g l_p \sin\phi - b\dot{\phi}\end{bmatrix} = \begin{bmatrix}U\\ V\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}\ddot{\theta}\\ \ddot{\phi}\end{bmatrix} = \frac{1}{\alpha\beta+\beta^2\sin^2\phi-\gamma^2\cos^2\phi}\cdot\begin{bmatrix}\beta & -\gamma\cos\phi\\ -\gamma\cos\phi & \alpha+\beta\sin^2\phi\end{bmatrix}\cdot\begin{bmatrix}U\\ V\end{bmatrix}$$

$$\begin{bmatrix}\ddot{\theta}\\ \ddot{\phi}\end{bmatrix} = \frac{1}{\alpha\beta+\beta^2\sin^2\phi-\gamma^2\cos^2\phi}\begin{bmatrix}\beta U - \gamma\cos\phi\cdot V\\ -\gamma\cos\phi\cdot U + (\alpha+\beta\sin^2\phi)V\end{bmatrix}$$

$$= \begin{bmatrix}f_1(\alpha,\beta,\gamma,\theta,\phi,\dot{\theta},\dot{\phi})\\ f_2(\alpha,\beta,\gamma,\theta,\phi,\dot{\theta},\dot{\phi})\end{bmatrix}$$

$$f_1' = \left\{ -\beta^2 \sin(2\phi)\,\dot{\theta}\dot{\phi} + \beta\gamma \sin\phi\,\dot{\phi}^2 + \beta(\tau - c\dot{\theta}) - \beta\gamma \sin\phi \cos^2\phi\,\dot{\theta}^2 + \gamma\cos\phi\left(-\tfrac{1}{2} M_p g l_p \sin\phi + b\dot{\phi}\right) \right\}$$

$$f_2' = +\beta\gamma \sin(2\phi)\cos(\phi)\,\dot{\theta}\dot{\phi} - \gamma^2 \sin\phi\cos\phi\,\dot{\phi}^2 - \gamma\cos\phi\,(\tau - c\dot{\theta}) + (\alpha + \beta\sin^2\phi)\left(\beta\sin\phi\cos\phi\,\dot{\theta}^2 - \tfrac{1}{2} M_p g l_p \sin\phi - b\dot{\phi}\right)$$

$$f_1 = f_1' \cdot \frac{1}{\alpha\beta + \beta^2\sin^2\phi - \gamma^2\cos^2\phi}$$

$$f_2 = f_2' \cdot \frac{1}{\alpha\beta + \beta^2\sin^2\phi - \gamma^2\cos^2\phi}$$

$$\boxed{\tfrac{1}{2} M_p g l_p = \delta}$$

$$\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} x_2 \\ f_1 \\ x_4 \\ f_2 \end{bmatrix} = g(x)$$

Linearization Around ~~stab~~ Equilibrium Points

Equilibrium points: $\dot{x} = 0$

$x_2 = 0$, $f_1 = 0$, $x_4 = 0$, $f_2 = 0$ $\quad \tau = 0$

$\dot{\theta} = 0$ $\quad \dot{\phi} = 0$

$$f_1 = 0 \Rightarrow \gamma\cos\phi\left(\tfrac{1}{2} M_p g\, l_p \sin\phi\right) = 0$$

$$f_2 = 0 \Rightarrow (\alpha + \beta\sin^2\phi)\cdot\left(\tfrac{1}{2} M_p g\, l_p \sin\phi\right) = 0$$

$$\Rightarrow \boxed{\phi = k\pi} \quad k \in \mathbb{Z} \text{ are equilibrium points.}$$

Linearization Around $\phi = 0$. $\quad x_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, $\quad \phi = \pi \quad x_1 = \begin{bmatrix} 0 \\ 0 \\ \pi \\ 0 \end{bmatrix}$

$$A_{x_0} = \left.\frac{\partial g}{\partial x}\right|_{x_0} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & \frac{-\beta c}{\alpha\beta - \gamma^2} & \frac{\gamma\delta}{\alpha\beta - \gamma^2} & \frac{\beta\gamma b}{\alpha\beta - \gamma^2} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{c\gamma}{\alpha\beta - \gamma^2} & \frac{-\alpha\delta}{\alpha\beta - \gamma^2} & -\frac{\alpha b}{\alpha\beta - \gamma^2} \end{bmatrix}$$

$$A_{x_1} = \left.\frac{\partial g}{\partial x}\right|_{x_1} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & \frac{-\beta c}{\alpha\beta - \gamma^2} & \frac{\gamma\delta}{\alpha\beta - \gamma^2} & -\frac{\beta b\gamma}{\alpha\beta - \gamma^2} \\ 0 & 0 & 0 & 1 \\ 0 & -\frac{c\gamma}{\alpha\beta - \gamma^2} & \frac{\alpha\delta}{\alpha\beta - \gamma^2} & -\frac{\alpha b}{\alpha\beta - \gamma^2} \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ \beta \\ 0 \\ -\gamma\cos\phi \end{bmatrix} \cdot \frac{1}{\alpha\beta + \beta^2 \sin^2\phi - \gamma^2 \cos^2\phi}$$

$$Bx_0 = \begin{bmatrix} 0 \\ \beta/(\alpha\beta-\gamma^2) \\ 0 \\ -\gamma/(\alpha\beta-\gamma^2) \end{bmatrix} \qquad Bx_1 = \begin{bmatrix} 0 \\ \beta/(\alpha\beta-\gamma^2) \\ 0 \\ \gamma/(\alpha\beta-\gamma^2) \end{bmatrix}$$

$$\dot{x} = Ax_0 \cdot x + Bx_0 \cdot \tau \qquad \text{near } : x_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2n\pi \\ 0 \end{bmatrix}$$

$$\dot{x} = Ax_1 \cdot x + Bx_1 \cdot \tau \qquad \text{near } x_1 = \begin{bmatrix} 0 \\ 0 \\ \pi \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ (2n+1)\pi \\ 0 \end{bmatrix}$$

Transfer Functions

Finding tranfer function for general position is very hard. I have written transfer function around equilibrium points.

$$\frac{X(s)}{\tau(s)} = (sI - A)^{-1} B$$

Around x_0

$$\frac{X(s)}{\tau(s)} = (sI - Ax_0)^{-1} \cdot Bx_0$$

~~Around~~

$$= \frac{1}{a_0 + a_1 s + a_2 s^2 + a_3 s^3 + s^4} \begin{bmatrix} h_1(s) \\ h_2(s) \\ h_3(s) \\ h_4(s) \end{bmatrix}$$

$a_0, a_1, a_2, a_3, \rightarrow$ constants

$$s^4 + a_3 s^3 + a_2 s^2 + a_1 s + a_0 = \det(sI - Ax_0)$$

Around x_1

$$\frac{X(s)}{\tau(s)} = \frac{1}{s^4 + b_3 s^3 + b_2 s^2 + b_1 s + b_0} \begin{bmatrix} h_1'(s) \\ h_2'(s) \\ h_3'(s) \\ h_4'(s) \end{bmatrix}$$

$$b_0 + b_1 s + b_2 s + b_3 s^3 + s^4 = \det(sI - Ax_1)$$

$\boxed{\text{Around } \phi = \pi}$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & a & b & c \\ 0 & 0 & 0 & 1 \\ 0 & d & e & f \end{bmatrix}$$

$$(sI - A)^{-1} \cdot B$$

$$= \begin{bmatrix} 1/s & \frac{s^2 - fs - e}{s\Delta(s)} & \frac{bs - bf + ce}{s\cdot\Delta(s)} & \frac{cs+b}{s\,\Delta(s)} \\ 0 & \frac{s^2 - fs - e}{\Delta(s)} & \frac{bs - bf + ce}{\Delta(s)} & \frac{cs+b}{\Delta s} \\ 0 & \frac{d}{\Delta(s)} & \frac{s^2 - (a+f)s + af - cd}{\Delta(s)} & \frac{s-a}{\Delta(s)} \\ 0 & \frac{ds}{\Delta(s)} & \frac{es + bd - ae}{\Delta(s)} & \frac{s(s-a)}{\Delta(s)} \end{bmatrix} \cdot B \qquad B = \begin{bmatrix} 0 \\ g \\ 0 \\ h \end{bmatrix}$$

$$\Delta(s) = s^3 - (a+f)s^2 + (af - e - cd)\,s + (ae - bd)$$

$$= \begin{bmatrix} g\cdot\left(\frac{s^2 - fs - e}{s\,\Delta(s)}\right) + h\cdot\left(\frac{cs+b}{s\,\Delta(s)}\right) \\ g\cdot\left(\frac{s^2 - fs - e}{\Delta(s)}\right) + h\cdot\left(\frac{cs+b}{\Delta(s)}\right) \\ g\cdot\frac{d}{\Delta(s)} + h\cdot\frac{s-a}{\Delta(s)} \\ g\cdot\frac{d\cdot s}{\Delta(s)} + h\cdot\frac{s(s-a)}{\Delta(s)} \end{bmatrix}$$

$$\left\{ \begin{aligned} a &= \frac{-\beta\cdot c}{\alpha\beta - \gamma^2} \\ b &= \frac{\gamma\delta}{\alpha\beta - \gamma^2} \\ c &= \frac{-\gamma\cdot b}{\alpha\beta - \gamma^2} \\ d &= \frac{-c\cdot\gamma}{\alpha\beta - \gamma^2} \\ e &= \frac{\alpha\delta}{\alpha\beta - \gamma^2} \\ f &= -\frac{\alpha\cdot b}{\alpha\beta - \gamma^2} \end{aligned} \right\}$$

$$\frac{\Theta(s)}{\tau(s)} = g\cdot\frac{s^2 - fs - e}{\Delta(s)} + h\cdot\frac{cs+b}{\Delta(s)}$$

$$= \frac{g\cdot(s^2 - fs - e) + h\cdot(cs+b)}{s^3 - (a+f)s^2 + (af - e - cd)s + (ae - bd)}$$

$$\left\{ \begin{aligned} g &= \frac{\beta}{\alpha\beta - \gamma^2} \\ h &= \frac{\gamma}{\alpha\beta - \gamma^2} \end{aligned} \right\}$$

$$\frac{\Phi(s)}{\tau(s)} = \frac{g\cdot d + h(s-a)}{s^3 - (a+f)s^2 + (af - e - cd)s + (ae - bd)}$$

Around $\phi = \pi$

Around $\phi = 0$

$c' = -c \qquad d' = -d \qquad e' = -e \qquad h' = -h$

$$\frac{\theta(s)}{\tau(s)} = \frac{g(s^2 - fs + e) + h(-c\cdot s + b)}{s^3 - (a+f)s^2 + (af + e - cd)\,s + (-ae + bd)}$$

$$\frac{\phi(s)}{\tau(s)} = \frac{-g\cdot d - h(s-a)}{s^3 - (a+f)s^2 + (af + e - cd)\,s + (-ae + bd)}$$

Include Motor Model

1) We can use a simplified method.

Actually, $L_m \frac{di}{dt} = V_{in} - R_m\, i - K_b \cdot N \cdot \dot{\theta}$

$N \cdot \dot{\theta} \rightarrow$ motor shaft speed

$K_b \cdot N \cdot \dot{\theta} \rightarrow$ Back EMF

$R_m \rightarrow$ resistance, $L_m \rightarrow$ inductance

$K_t \rightarrow$ torque constant

$K_b \rightarrow$ back-emf constant

$N \rightarrow$ gear ratio

$V_{in} \rightarrow$ input voltage.

If we assume $L_m/R_m <<$ sample time, we can use quasi-static motor model.

$$L_m \cdot \frac{di}{dt} \simeq 0.$$

$$i = \frac{V_{in} - K_b \cdot N \cdot \dot{\theta}}{R_m}$$

$$\tau = N \cdot K_t \cdot \left(\frac{V_{in} - K_b \cdot N \cdot \dot{\theta}}{R_m} \right)$$

2) We can use full state

$$x = \begin{bmatrix} \text{theta} \\ \text{theta\_dot} \\ \text{phi} \\ \text{phi\_dot} \\ \text{current} \end{bmatrix} = \begin{bmatrix} \theta \\ \dot{\theta} \\ \phi \\ \dot{\phi} \\ i \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1/L_m \end{bmatrix}$$

$$\dot{x} = \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \\ \dot{\phi} \\ \ddot{\phi} \\ \dot{i} \end{bmatrix} = \begin{bmatrix} \dot{\theta} \\ f_1 \\ \dot{\phi} \\ f_2 \\ f_3 \end{bmatrix}$$

$$\text{Den} = \alpha\beta + \beta^2 \sin^2\phi - \gamma^2 \cos^2\phi$$

$$f_1 = \frac{1}{\text{Den}} \Big[-\beta^2 \sin(2\phi)\, \dot{\theta}\dot{\phi} + \beta\gamma \sin\phi \cdot \dot{\phi}^2 + \beta(N \cdot K_t \cdot i - c \cdot \dot{\theta}) - \beta\gamma \sin\phi \cdot \cos^2\phi \cdot \dot{\theta}^2 + \gamma \cos\phi \left(\tfrac{1}{2} M_p\, g\, l_p \sin\phi\right) + \gamma \cos\phi \cdot b \cdot \dot{\phi} \Big]$$

$$f_2 = \frac{1}{Den}\Big[\beta\gamma \sin(2\phi)\cos(\phi)\,\dot{\theta}\,\dot{\phi} - \gamma^2 \sin\phi \cos\phi\,\dot{\phi}^2$$
$$- \gamma \cos\phi\,(N.K_t.i - c\,\dot{\theta})$$
$$+ (\alpha + \beta\sin^2\phi)\,(\beta \sin\phi\,\cos\phi\,\dot{\theta}^2$$
$$- \tfrac{1}{2} M_p\, g\, l_p \sin\phi - b\,\dot{\phi})\Big]$$

$$f_3 = (V_{in} - R_m . i - K_b . N . \dot{\theta}) / L_m .$$

$V_{in} \rightarrow$ is input here.